

Leo Beck

Computational Materials Scientist | Scientific ML
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EDUCATION

University of Colorado Boulder

Doctor of Philosophy in Materials Science & Engineering Aug 2023 – Present
Master of Science in Materials Science & Engineering GPA: 3.90/4.00

Clemson University

Bachelor of Science in Materials Science & Engineering Aug 2019 - May 2023
Bachelor of Science in Mathematical Sciences GPA: 3.84/4.00

EXPERIENCE

Graduate Researcher, *Heinz Interfaces Lab – University of Colorado Boulder* Dec 2023 – Present

- Develop supervised ML models from heterogeneous, class-imbalanced materials datasets using feature engineering, resampling, and model validation
- Automate high-throughput workflows for model setup, simulation, and analysis using Python, Bash, Packmol, and C++
- Apply LAMMPS simulations and time-series analysis to quantify structural evolution in perovskites

High-Performance Computing Intern, *Air Force Research Laboratory* Summers 2024 & 2026

- Performed replicated LAMMPS simulations to characterize shear strength and thermal conductivity of MXenes
- Calculated slip probability vs. force and fit logistic curves to extract inter-sheet shear strength
- Automated structure creation, post-processing, and visualization in Python

Classical Simulations Consultant, *FAIRmat* Summer 2025

- Extended the NOMAD LAMMPS parser in Python
- Curated regression tests and improved parser robustness across edge cases; documented assumptions and failure modes and advised on LAMMPS methods/options

Engineering and Planning Intern, *Benore Logistic Systems, Inc.* Summer 2023

- Built driver-utilization visualization tool & automated daily analytics with VBA and Power Automate
- Reduced manual planning and load-splitting steps by implementing a first-fit decreasing algorithm to automate pallet-to-truck packing in VBA

SELECTED PUBLICATIONS

M Karabin, I Armstrong, **L Beck**, et al. (2026) Enhancing Dimensionality Prediction in Hybrid Metal Halides via Feature Engineering and Class-Imbalance Mitigation

I Armstrong, J Slocik, **L Beck**, et al. (2026) Validated Reactive Force Field Quantifies MXene Interfacial Properties, Mechanics, and Thermal Transport

TECHNICAL SKILLS

Machine Learning: PyTorch, statistical modeling, feature engineering, model validation, optimization
Programming: Python, C/C++, MATLAB, Bash, Git, Linux
Molecular Simulation: LAMMPS, NAMD, VMD, Materials Studio, Packmol, VASP, ORCA
Data & Automation: Scientific data analysis, scikit-learn, Excel, Power Automate, VBA

LEADERSHIP & TEACHING

Physics Teaching Assistant Fall 2023, Spring 2026

- Led recitations of Physics 1 and Physics 2 lab. Graded Classical Mechanics 1

Financial and Logistics Officer, *Clemson Formula SAE* May 2021 - Jun 2022

- Increased annual team funding from \$30,000 to \$70,000; managed a \$100,000 budget and logistics of 30 members and coordinated purchasing across 8 engineering divisions